## *1.6.2 Orchestration*

So you have decided to use containers, great! You can rely on their portability to deploy them to any infrastructure providing a container runtime. You can achieve reproducibility, so there are no bad surprises when moving containers from development to staging to production. You can scale them quickly, since they're so lightweight, and get high availability for your applications. You're ready to adopt them for your next cloud native system. Or are you?

Provisioning and managing containers on a single machine is pretty straightforward. But when you start dealing with tens or hundreds of containers scaled and deployed on several machines, you need something else.

When you move from virtual servers (the IaaS model) to container clusters (the CaaS model), you're also switching your point of view.[15] In IaaS you focus on single computing nodes, which are virtual servers. In CaaS the underlying infrastructure is abstracted, and you're focusing on clusters of nodes.

With the new perspective offered by CaaS solutions, the deployment target will not be a machine anymore but rather a cluster of machines. CaaS platforms, such as those based on Kubernetes, provide many features to address all the significant concerns we look for in a cloud native environment, *orchestrating* containers across machines. The two different topologies are shown in figure 1.11.